Employee Payroll Data Cleaning & Financial Analysis project.

Step 1: Dataset Explanation

- **Employee ID** → Must be unique (primary identifier).
- **Department** → Categorical, often used for grouping/aggregation.
- **Basic Salary** → Core fixed pay.
- **Bonus** → Variable pay, may differ by department/role.
- **Deductions** → Taxes, PF, insurance, etc.
- **Net Salary** → Final payout (Gross – Deductions).

Derived fields:

- **Gross Salary = Basic Salary + Bonus**
- **Net Salary = Gross Salary – Deductions**

Step 2: Data Cleaning Checklist

Think like a Business Analyst ensuring financial accuracy:

- **Employee ID**
  - Check for duplicates → no two employees should share the same ID.
  - Validate format (numeric/alphanumeric consistency).
- **Department**
  - Standardize names (e.g., "HR" vs "Human Resources").
  - Handle missing values (assign "Unknown" or validate with HR records).
- **Salary Fields**
  - Ensure numeric format (no text, currency symbols).
  - Detect outliers (e.g., negative salaries, unrealistic bonuses).
  - Cross-check: Net Salary must equal Gross – Deductions.
- **Business Rules Validation**
  - Deductions should not exceed Gross Salary.
  - Bonus should not be negative.
  - Net Salary must always be ≥ 0.

Step 3: Business Understanding

Why this matters in HR/Finance:

- **Salary disputes** → arise if Net Salary doesn't match business rules.
- **Compliance issues** → incorrect deductions can break tax laws.
- **Audit failures** → mismatched records trigger red flags.

So, before dashboards, you must:

- Confirm **business logic** (how bonuses/deductions are calculated).
- Align with **HR/Finance expectations** (e.g., PF %, tax slabs).
- Document assumptions clearly.

Step 4: Structuring the Data

Prepare for analysis:

- Create a **clean master table** with validated fields.
- Add calculated columns (Gross Salary, Net Salary).
- Ensure consistent data types (numeric, categorical).
- Build a **data dictionary** explaining each column and its business meaning.

✅ By the end of Week 2, your deliverable should be:

1. **Clean dataset** (no duplicates, errors, or inconsistencies).
2. **Data dictionary** (clear explanation of each field).
3. **Business logic document** (rules for salary calculation).

Dataset Understanding (Finance Lens)

- **Employee ID → Must be Unique**
  - Serves as the **primary key**.
  - No duplicates allowed; otherwise, salary records could overlap or cause disputes.
  - Missing IDs are unacceptable—every employee must be traceable.
- **Gross Salary → Before Deductions**
  - Formula: **Gross Salary = Basic Salary + Bonus**
  - Represents the total earnings before statutory or voluntary deductions.
  - Must always be ≥ **Basic Salary** (since bonus adds value, not subtracts).
  - Outlier check: unusually high bonuses compared to basic salary may indicate data entry errors.
- **Net Salary → Final Payout**
  - Formula: **Net Salary = Gross Salary – Deductions**
  - This is the actual amount credited to employees.
  - Must always be ≥ **0** (negative payouts are invalid).
  - Cross-check: Net Salary should match payroll slips and compliance records.

Business Logic Validation

Before cleaning, confirm:

- **Deductions** follow HR/Finance rules (tax %, PF %, insurance caps).
- **Bonus** policies are consistent across departments.
- **Net Salary** aligns with compliance (no underpayment or miscalculation).

Cleaning Priorities

- Remove duplicate Employee IDs.
- Standardize Department names.
- Ensure salary fields are numeric and positive.
- Validate formulas:

- Gross Salary = Basic + Bonus
- Net Salary = Gross – Deductions

By the end of this stage, you should have:

1. **Validated dataset** (no duplicates, consistent salary logic).
2. **Business rules documented** (how deductions/bonuses are applied).
3. **Data dictionary updated** (clear explanation of each column + formula).

Key Payroll Analysis Questions

1. **Which department has the highest total payroll cost?**
   - Aggregate **Net Salary** by department.
   - Helps identify cost-heavy departments for budgeting and HR planning.
2. **Which department has the highest average salary?**
   - Calculate **mean Net Salary** per department.
   - Useful for fairness checks and compensation benchmarking.
3. **Are there any employees receiving unusually high or low salaries?**
   - Detect **outliers** using statistical methods (e.g., salaries 2–3 standard deviations from the mean).
   - Prevents disputes and flags potential data entry errors.
4. **Which department contributes the highest tax deductions?**
   - Sum **Deductions** by department.
   - Important for compliance and understanding tax burden distribution.
5. **Are there duplicate employee records affecting payroll?**
   - Check for duplicate **Employee IDs**.
   - Critical to avoid double payments or audit failures.
6. **What percentage of total payroll goes toward bonuses?**
   - Formula: **(Total Bonus ÷ Total Gross Salary) × 100**.
   - Reveals incentive structure and its impact on payroll.
7. **Is the net salary correctly calculated for all employees?**
   - Validate: **Net Salary = Gross Salary – Deductions**.
   - Ensures accuracy before dashboards or reporting.
8. **Which department has the highest deduction ratio?**
   - Formula: **Total Deductions ÷ Gross Salary per department**.
   - Shows how much of earnings are reduced by deductions, useful for compliance and employee satisfaction.

Step 2: Cleaning Tasks

1. **Remove Duplicate Employees**
   - Check for duplicate **Employee IDs**.
   - If duplicates exist → salary may be paid twice.
   - Action: Keep only one record per employee, verify with HR master list.
2. **Check Incorrect Salary Values**
   - Validate **Basic Salary, Bonus, Deductions** are numeric and non-negative.
   - Flag outliers (e.g., Basic Salary = 0, Bonus > Basic Salary × 3).
   - Action: Correct or escalate anomalies to HR/Finance.
3. **Standardize Department Names**

    - Ensure consistent naming (e.g., “HR” vs “Human Resources”).
    - Missing department → map correctly or assign “Unknown”.
    - Action: Create a **department mapping table** for standardization.
4. **Validate Deduction Formulas**
    - Confirm **Net Salary = Gross Salary – Deductions**.
    - Deductions must not exceed Gross Salary.
    - Action: Recalculate Net Salary if mismatches are found.

Cleaning Logic (Finance Perspective)

- **Duplicate Employees → Salary Paid Twice** → Leads to payroll inflation and audit failures.
- **Incorrect Tax Calculation** → Compliance risk, penalties, and employee disputes.
- **Missing Department Mapping** → Breaks departmental cost analysis and reporting.

Step 3: Business Analysis KPIs

KPI	Definition	Business Purpose	Formula / Calculation
Total Payroll Cost	Overall company salary expenditure	Tracks total salary burden for budgeting	Sum of all **Net Salaries**
Average Salary per Department	Mean salary per department	Benchmarks pay levels across teams	Avg(Net Salary) grouped by Department
Highest Paid Department	Department with largest payroll spend	Identifies cost concentration	Max (Sum(Net Salary) by Department)
Total Deductions	Sum of all statutory & tax deductions	Compliance and tax tracking	Sum(Deductions)
Bonus Percentage of Payroll	Share of payroll spent on bonuses	Evaluates incentive structure	(Sum(Bonus) ÷ Sum(Gross Salary)) × 100
Net vs Gross Salary Ratio	Ratio of final payout vs gross	Measures deduction impact	Net Salary ÷ Gross Salary
Payroll Growth Rate	Change in payroll cost over time	Indicator of workforce expansion	((Current Payroll – Previous Payroll) ÷ Previous Payroll) × 100
Overtime Cost Percentage	Share of payroll spent on overtime	Tracks operational efficiency	(Sum(Overtime Pay) ÷ Total Payroll) × 100
Salary Variance Across Departments	Spread of salaries between departments	Pay equity analysis	Variance(Net Salary by Department)
Duplicate Payroll Entries	Count of duplicate employee records	Financial risk control	Count of duplicate Employee IDs

Why These KPIs Matter

- **Finance perspective** → ensures compliance, cost control, and audit readiness.
- **HR perspective** → highlights fairness, incentives, and workforce trends.

- **Business perspective** → connects payroll data to strategic decisions (growth, efficiency, equity).

Step 4: Payroll KPI List

KPI	Definition	Business Purpose	Formula / Calculation
Total Deductions	Sum of all statutory/tax deductions	Compliance & tax tracking	Sum(Deductions)
Net Payroll Cost	Total payout after deductions	Actual company salary expenditure	Sum(Net Salary)
Average Salary per Department	Mean salary per department	Pay benchmarking & fairness	Avg(Net Salary) grouped by Department
Highest Paid Department	Department with largest payroll spend	Identifies cost concentration	Max(Sum(Net Salary) by Department)
Bonus Percentage of Payroll	Share of payroll spent on bonuses	Incentive structure analysis	(Sum(Bonus) ÷ Sum(Gross Salary)) × 100
Salary Variance Across Departments	Spread of salaries between departments	Pay equity analysis	Variance(Net Salary by Department)
Overtime Cost Percentage	Share of payroll spent on overtime	Operational efficiency tracking	(Sum(Overtime Pay) ÷ Total Payroll) × 100
Payroll Growth Rate	Change in payroll cost over time	Workforce expansion indicator	((Current Payroll – Previous Payroll) ÷ Previous Payroll) × 100
Deduction Ratio	Ratio of deductions to gross salary	Measures deduction impact	Deductions ÷ Gross Salary
Duplicate Payroll Records Count	Number of duplicate employee IDs	Financial risk control	Count of duplicate Employee IDs
Tax Contribution by Department	Department-wise tax/deduction share	Compliance & audit readiness	Sum(Deductions) grouped by Department

Why This KPI List Matters

- **Finance** → ensures compliance, tax accuracy, and cost control.
- **HR** → highlights fairness, incentives, and workforce trends.
- **Business** → connects payroll data to strategic decisions (growth, efficiency, equity).

Step 5: Business Insight

Finance teams use payroll KPIs not just for reporting, but for **strategic decision-making**. Here's how each KPI connects to business impact:

- **Total Deductions** → Ensures compliance with tax laws and statutory contributions. Helps track whether deductions align with legal requirements.
- **Net Payroll Cost** → Shows the company's true salary expenditure, critical for budgeting and cash flow planning.
- **Average Salary per Department** → Benchmarks compensation fairness and identifies departments with higher pay structures.
- **Highest Paid Department** → Highlights cost concentration, guiding resource allocation and workforce planning.
- **Bonus Percentage of Payroll** → Reveals incentive structures, helping HR/Finance evaluate if bonuses are sustainable or motivating.
- **Salary Variance Across Departments** → Detects pay equity issues, supporting fair compensation policies.
- **Overtime Cost Percentage** → Measures operational efficiency; high overtime costs may signal understaffing or poor workload distribution.
- **Payroll Growth Rate** → Indicates workforce expansion or rising salary costs, useful for long-term financial forecasting.
- **Deduction Ratio (Deductions ÷ Gross Salary)** → Shows how much of employee earnings are reduced by deductions, impacting employee satisfaction and compliance.
- **Duplicate Payroll Records Count** → Identifies financial risks like double payments, preventing audit failures.
- **Tax Contribution by Department** → Breaks down tax burden by department, useful for compliance audits and financial transparency.

Strategic Value

- **Cost Management** → Finance teams can identify where payroll costs are concentrated and optimize budgets.
- **Compliance Assurance** → Validating deductions and tax contributions reduces risk of penalties.
- **Hiring & Workforce Planning** → Payroll growth, overtime costs, and departmental salary benchmarks guide hiring decisions.
- **Equity & Transparency** → Salary variance and deduction ratios ensure fairness and build employee trust.

By the end of Step 5, you should have:

1. **Business Insight Report** → A narrative explaining how each KPI informs Finance strategy.
2. **Clean dataset + KPI calculations** → Ready for Week 3 dashboards.
3. **Stakeholder-ready documentation** → Showing not just numbers, but the *story behind the payroll*.